



Validation Frequency - CNN Training

Follow

60 views (last 30 days)

Andrea Daou on 18 Jan 2021

Moved: Arkadiy Turevskiy **STAFF** on 11 Jan 2024

Accepted Answer: Ullah Nadeem **LEVEL 1**

👍 Vote | 0

🔗 Link

Hello,

I have a question concerning the **validation frequency** in options when fine-tuning a pretrained network (image classification problem) :

```
opts = trainingOptions('sgdm', 'InitialLearnRate', 0.001,...  
  
    'MaxEpochs', 10, 'MiniBatchSize', 32,...  
  
    'Shuffle','every-epoch','Verbose',true, ...  
  
    'ValidationData',augmentedTestSet, ...  
  
    'ValidationFrequency','valFrequency', ...  
  
    'Plots','training-progress');
```

mynet = trainNetwork(augmentedTrainingSet, lgraph, opts);

How can I choose the best valFrequency value for my model ?

Thank you !!

0 Comments

Sign in to comment.

Sign in to answer this question.

Accepted Answer

MATLAB Answers

Ullah Nadeem **LEVEL 1** on 20 Oct 2022

Moved: Arkadiy Turevskiy **STAFF** on 11 Jan 2024

👍 Vote | 0

🔗 Link

Hello Andrea!

VF: Validation Frequency

As far as I know, the selection of validation frequency depends on the following:

(1) The amount of training data

If the training data is large and the VF if small number (let's say after 30 iterations), the model will not learn enough while validating more oftenly results in long training time and maybe stop the training early if the 'ValidationPatience' is not set to 'Inf'.

(2) How fast the model learns the data well

If a problem is easy and the model learns it quickly, the VF supposed to be in range of 1/15th times of the iterations/epoch to 1/7th times of the iterations/epoch to let the network stop early and not gets overfit. But if the problem is complicated and model can't learns it quickly, then it is better to select the VF in range of 1/7th times of the iterations/epoch to 1/3th times of the iterations/epoch to let the model learns enough before validating.

Note: Iterations/epoch can be calculated as total number of training samples/mini-batch size.

0 Comments

Sign in to comment.

More Answers (0)

Sign in to answer this question.

See Also

MATLAB Answers

How to determine the correct number of Validation Frequency during neural network training
1 Answer

How can I add new images to a trained deep learning network to classify new images ?
1 Answer

How is a deep learning "training accuracy" calculated?
0 Answers

Entire Website

Image classification using CNN with mixup data augmentation
File Exchange

Image Classification using CNN with Multi Input 複数の入力層を持つCNN
File Exchange

Image Classification for Non-Data Scientists
File Exchange

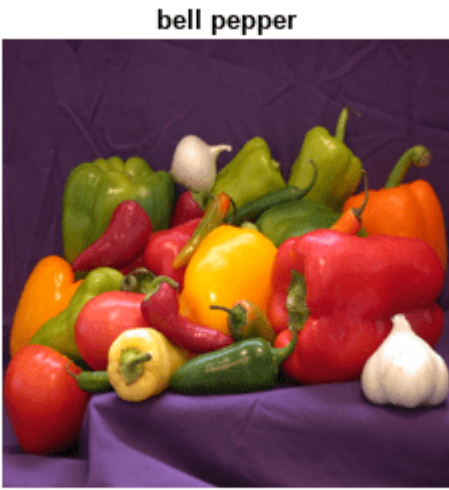
Categories

AI, Data Science, and Statistics > Deep Learning Toolbox

Find more on **Deep Learning Toolbox** in Help Center and File Exchange

Tags

- classification
- image
- neural network
- neural networks



Classify an Image Using a Pretrained Network

» Try this example

Poll

My favorite sport to watch is

- ☐ American style football
- ☐ Soccer / football
- ☐ baseball
- ☐ basketball
- ☐ tennis or golf
- ☐ rugby, track, cricket, racing, etc.

3679 votes 16 comments 407 views

Vote

mathworks.com

© 1994-2024 The MathWorks, Inc. MATLAB and Simulink are registered trademarks of The MathWorks, Inc. See mathworks.com/trademarks for a list of additional trademarks. Other product or brand names may be trademarks or registered trademarks of their respective holders.

Join the conversation